

ApexPlanet Data Analytics Internship

Final Project Report

Student: Amit

Dataset: Supermarket Sales Dataset

Project Type: End-to-End Data Analytics

1. Project Overview

This project was completed as part of the ApexPlanet Data Analytics Internship. The objective was to perform an end-to-end analysis of supermarket sales data, starting from data cleaning and exploratory analysis and progressing through SQL analysis, visualization, Power BI dashboarding, statistical analysis, customer segmentation, and machine learning.

2. Project Objectives

- Understand the structure and quality of the supermarket sales dataset.
- Clean and preprocess data for reliable analysis.
- Extract business insights using SQL and Python.
- Create meaningful charts and an interactive Power BI dashboard.
- Apply statistics, customer segmentation, and predictive modelling techniques.
- Present findings in a clear and business-oriented format.

3. Tasks Completed

Task	Work Completed
Task 1 – Exploratory Data Analysis	Performed data inspection, cleaning, data-type handling, distributions, correlation analysis, sales patterns and exploratory analysis.
Task 2 – SQL Analysis	Created a SQLite database and performed SQL-based filtering, aggregation, grouping, joins and business insights.
Task 3 – Visualization & Power BI	Built charts using Python libraries and created an interactive Power BI dashboard containing KPIs, trends and insights.
Task 4 – Advanced Analytics	Applied descriptive statistics, correlation analysis, hypothesis testing, time-series analysis, K-Means clustering and regression models.
Task 5 – Final Documentation	Organized the project files, reports, visualizations, notebooks and documentation into a reusable project template.

4. Tools & Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, SciPy, Scikit-learn, Statsmodels, Jupyter Notebook, SQL/SQLite and Microsoft Power BI.

5. Dataset

The project uses the Supermarket Sales Dataset. The dataset contains transaction-level information such as branch, city, customer type, gender, product line, unit price, quantity, tax, total sales, payment method, customer rating and related sales attributes.

6. Analysis Performed

Exploratory analysis was used to identify distributions, relationships, trends and unusual observations. SQL was used to query the cleaned data through a database. Visualizations were prepared to communicate sales performance, product-line contribution, customer characteristics, payment methods, branch/city performance and relationships among numerical variables.

Advanced analysis included statistical summaries, correlation analysis, hypothesis testing, time-based sales analysis, K-Means clustering for customer segmentation, and predictive modelling using machine-learning algorithms available in the project notebooks.

7. Key Business Insights

- Sales and gross income show a positive relationship.
- Product lines contribute differently to overall revenue and sales volume.
- Customer characteristics such as customer type and gender can be compared to understand purchasing patterns.
- Branch and city-level analysis helps identify differences in sales performance.
- Customer segmentation can separate customers into groups with different purchasing behaviour and value.
- Interactive dashboards make the analysis easier to explore and support business decision-making.

8. Project Structure

The GitHub project is organized into folders for raw and processed data, notebooks, SQL scripts/database files, visualizations, Power BI dashboard assets, reports and supporting scripts. This structure makes the project easier to understand, reproduce and review.

9. Conclusion

This project demonstrates a complete data analytics workflow using a real-world style supermarket sales dataset. The work combines data preparation, exploratory analysis, SQL, visualization, dashboarding, statistical methods, clustering and machine learning. Completing the project strengthened practical skills in Python-based analytics and helped translate raw transaction data into actionable business insights.

10. Future Improvements

Future work could include automated data-refresh pipelines, more advanced forecasting, additional customer-level features, model evaluation and tuning, deployment of the dashboard or analysis as a web application, and scheduled reporting.

Prepared for GitHub submission

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